

Build a Document Question Answering (Document QA) Agent

Question 1 (from document)

The screenshot shows the workflow editor for 'My workflow 4'. The workflow consists of four steps: 'When chat message received', 'Read/Write Files from Disk', 'Extract from File', and 'Basic LLM Chain'. The 'Basic LLM Chain' step is connected to a 'Groq Chat Model'.

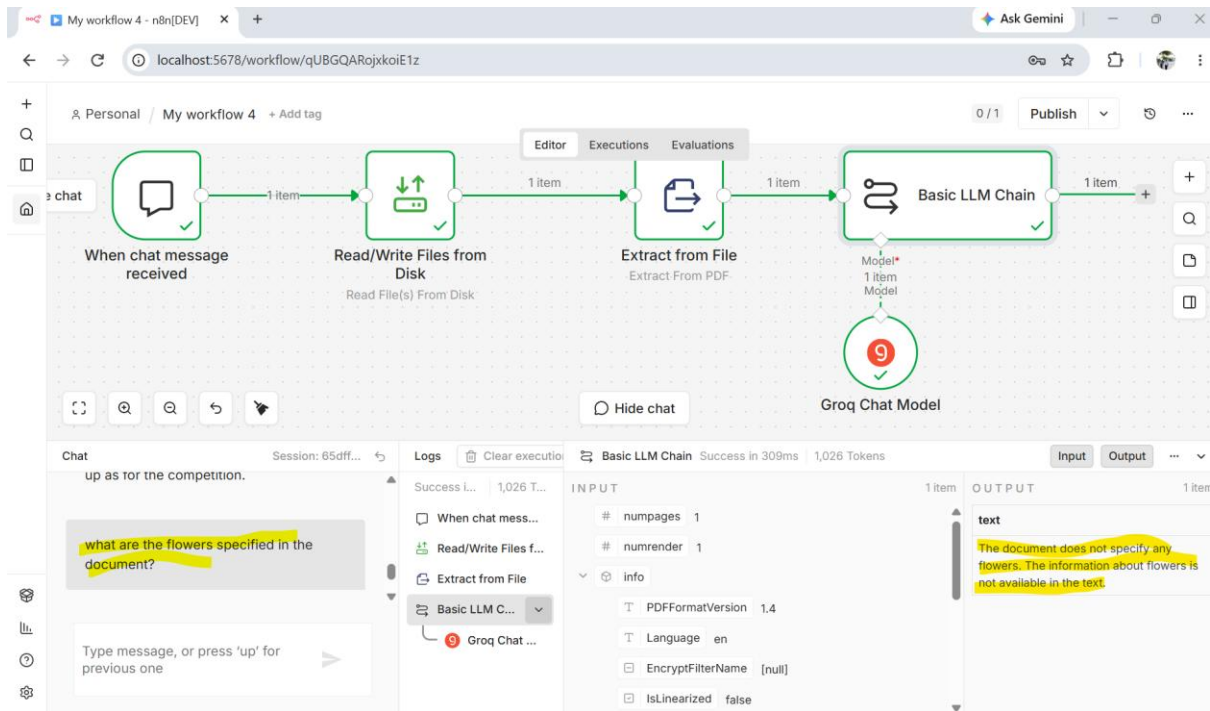
The chat interface shows the user's question: "what are the materials required?". The output of the 'Basic LLM Chain' step is displayed in the 'OUTPUT' tab, showing the text: "The materials required for the Inter-House Competition 2 are: 1. Costume of any Indian Freedom Fighter. 2. Relevant accessories/props (optional and safe to carry). 3. A well-prepared 1-minute speech ending with a meaningful slogan."

Question 2 (from document)

The screenshot shows the workflow editor for 'My workflow 4'. The workflow consists of four steps: 'When chat message received', 'Read/Write Files from Disk', 'Extract from File', and 'Basic LLM Chain'. The 'Basic LLM Chain' step is connected to a 'Groq Chat Model'.

The chat interface shows the user's question: "what are the freedom fighters example names given in the document". The output of the 'Basic LLM Chain' step is displayed in the 'OUTPUT' tab, showing the text: "The example names of freedom fighters given in the document are: 1. Mahatma Gandhi 2. Bhagat Singh 3. Rani Lakshmbai 4. Subhas Chandra Bose 5. Sarojini Naidu 6. Dr. B.R. Ambedkar 7. Chandrashekar Azad These are the specific names mentioned in the document as examples of Indian freedom fighters that students can"

Question 3 (outside of document)



Overview of the Workflow

This n8n workflow functions as an AI-powered Document Assistant and Question-Answering Pipeline. It allows users to interact via a chat interface and ask questions about a specific local document (such as a school circular or event notice like *Grade3_IHC2 circular_Freedom Fighters as Water Warriors*). The workflow automatically reads the file, extracts its text, and feeds both the document content and the user's question into a Large Language Model to generate precise, grounded answers.

Node-by-Node Functionality

- 1. When chat message received (Chat Trigger)
 - Role: The entry point for user interaction.
 - Function: Listens for incoming messages typed by the user in the embedded n8n chat window (e.g., "When is the competition happening?" or "What are the exam details?").
- 2. Read/Write Files from Disk
 - Role: File Ingestion Layer.
 - Function: Accesses the local file system or directory to load the target document file (.docx or .pdf) as a binary data buffer, making its raw contents available to the rest of the workflow.
- 3. Extract from File
 - Role: Document Parsing Layer.

- Function: Takes the binary file buffer from the disk node and parses it into clean, readable text content, stripping away formatting metadata so that downstream AI models can process the text easily.
 - 4. Basic LLM Chain & Groq Chat Model
 - Role: The AI Reasoning & Generation Engine.
 - Function: Combines the user's chat question with the extracted document text as context. Powered by the Groq Chat Model, this node analyzes the text and formulates a direct, fact-checked response sourced exclusively from the uploaded file.
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How It Arrives at the Output (Execution Lifecycle)

1. User Input: A user types a question into the chat widget (e.g., *"What are the names of freedom fighters mentioned?"*).
2. File Retrieval: The Read/Write Files from Disk node fetches the source document from the local folder.
3. Text Extraction: The Extract from File node converts the document into readable text format.
4. AI Contextual Analysis: The Basic LLM Chain receives both the user's chat prompt and the full document text. It matches the query against the document contents.
5. Response Delivery: The model generates the final, accurate answer and streams it back to the chat interface in real time.

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